Faiyaj Rahman

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Education

University of Michigan — Ann Arbor, MI
B.S.E. in Computer Engineering — *May 2028*
- **Relevant Coursework:** Data Structures and Algorithms, Computer Organization, Object Oriented Programming, Discrete Math, Computing Systems, Computational Linear Algebra, Vector Calculus

Honors & Awards

Jane Street FOCUS Participant — May 2026
*Competitive quantitative trading and software engineering program, selected from **1000+** applicants* — *New York, NY*

Work Experience

Software Engineering Intern — May 2026 – Aug. 2026
Bosch | Agentic AI, Python, AWS, RESTAPIs, CI/CD, Docker, Kubernetes, Bash, Git — *Farmington Hills, MI*
- Deployed an operations platform to AWS and scripted multi-tenant KPIs and analytics scaling to **5000+** vehicles.
- Designed a tool to convert CAN `.asc` logs to `MF4` with DBC channel mapping, saving **20+** engineer-hours monthly.
- Shipped a 3-agent pipeline, cleaning and tagging internal docs for a master agent, cutting new-hire escalations **40%**.
- Configured a robust RESTAPI data pipeline, boosting data extraction speeds for project management analytics by **5x**.
- Engineered a multi-threaded GUI tool used by **100+** developers to build Python executables, saving **$15K** in licenses.
- Scoped multiple user stories in **2**-week sprints with code reviews, demoing **4** tools org-wide to non-technical customers.

Software Developer — Nov. 2025 – Present
Saf · Darul Uloom Michigan | Claude, TypeScript, React, Supabase, PostgreSQL, Tailwind, Git — *Warren, MI*
- Launched a school-management beta on Postgres, an Express REST API, and React with role-based access control.
- Implemented assignment making, grading, attendance, and an admin panel, replacing paper grades for **20+** educators.
- Ran the school's live pilot for **3** months in 2-week sprints, shipping **5** iterations from educator review with feedback.
- Currently contributing to the production platform and implementing AI feature work ahead of scholar-backed launch.

Software Engineer — Aug. 2025 – Present
MRacing FSAE | Cursor, C/C++, Embedded Systems/Software, Linux, CAN, Raspberry Pi, Gitlab — *Ann Arbor, MI*
- Built a C++ Raspberry Pi dashboard decoding **1000+** CAN messages/sec into signals and fault states across **3** pages.
- Wrote firmware for an IR lap trigger, edge-detecting photosensor voltage drops on beam break to timestamp laps.
- Partnered with the testing subteam to debug and validate software on-vehicle, ensuring ~**100%** data consistency.

Undergraduate Research Software Developer — Sep. 2025 – May 2026
University of Michigan Electric Vehicle Center | Claude, Python, Raspberry Pi, Figma, Simulink — *Ann Arbor, MI*
- Developed open-source Python software for an EV charging module, contributing to **$10k** in net savings for an OEM.
- Designed a Simulink FSM for system-safety logic and charging protocols, porting it to Python for pipeline integration.

Projects

Overround | *C++, Python, PyTorch, React, D3.js, FastAPI, PostgreSQL* — Jul. 2026 – Present
- Building a transformer with learned club embeddings over **18,000+** matches across five domestic leagues, solving cross-league calibration to predict UEFA Champions League outcomes, outperforming Dixon-Coles and Elo baselines.
- Engineering a multithreaded C++ Monte Carlo engine simulating the **36**-team league phase at **5,000+** sims per sec.

Limit Order Book & Matching Engine | *Claude, C++, Linux, Git* — May 2026 – Present
- Building a C++ limit order book with price–time priority at **10K–100K+** orders/sec and **sub-millisecond** latency.
- Optimized with cache-efficient data structures and custom allocators, cutting latency ~**60%** over **1M+** order events.

Low-Level Modulator | *Verilog, Assembly, FPGA, Digital Logic* — Jan. 2026 – Apr. 2026
- Engineered a **18**-instruction processor in Verilog on a DE2-115 FPGA driving a synthesizer with **40** assembly modules.
- Designed an integer-math DSP pipeline synthesizing **32-bit** audio from **3** wavetables with **5** modulations at **30 FPS**.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, SQL, Assembly, Verilog, MATLAB
Tools/Frameworks: Claude, Cursor, Codex, React, Node, FastAPI, PostgreSQL, AWS, Docker, Linux, Raspberry Pi
Practices: Agile/Scrum, Git/GitHub, CI/CD, code review, unit testing, REST API design, RAG & LLM agents